A short guide to RAG

Retrieval-augmented generation

RAG helps a language model answer questions using your documents.

It has three main steps:

1. Retrieve relevant passages

The system searches the document index for text related to your question.

Embeddings represent meaning as numerical vectors for similarity search.

2. Build context

The retrieved passages are included alongside your question in the prompt.

Only a small number of useful passages need to be sent to the model.

3. Generate a grounded answer

The local language model writes an answer using the supplied context.

Source references let you check the original document and page.

This application uses PyPDF, LangChain, ChromaDB, and a local model server.

Both Ollama and LM Studio can supply local models.